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MODBUS on Ethernet TCP/IP

Protocol Reference Guide

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This document applies to the MODBUS on Ethernet TCP/IP.

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Preface

Preface Sections:

- Purpose of this Guide
- Special Notation

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Purpose of this Guide

This manual provides information on the MODBUS on TCP/IP solution for SST-CLX Backplane modules. The MODBUS solution creates a hierarchical structure, enabling one or more smart slaves to be consulted by a single master.

Special Notation

The following special notations may be used throughout this guide:



Warning

Warning messages alert the reader to situations where personal injury may result. Warnings are accompanied by the symbol shown, and precede the topic to which they refer.



Caution

Caution messages alert the reader to situations where equipment damage may result. Cautions are accompanied by the symbol shown, and precede the topic to which they refer.



Note

A note provides additional information, emphasizes a point, or gives a tip for easier operation. Notes are accompanied by the symbol shown, and follow the text to which they refer.

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1 General Information

Chapter Sections:

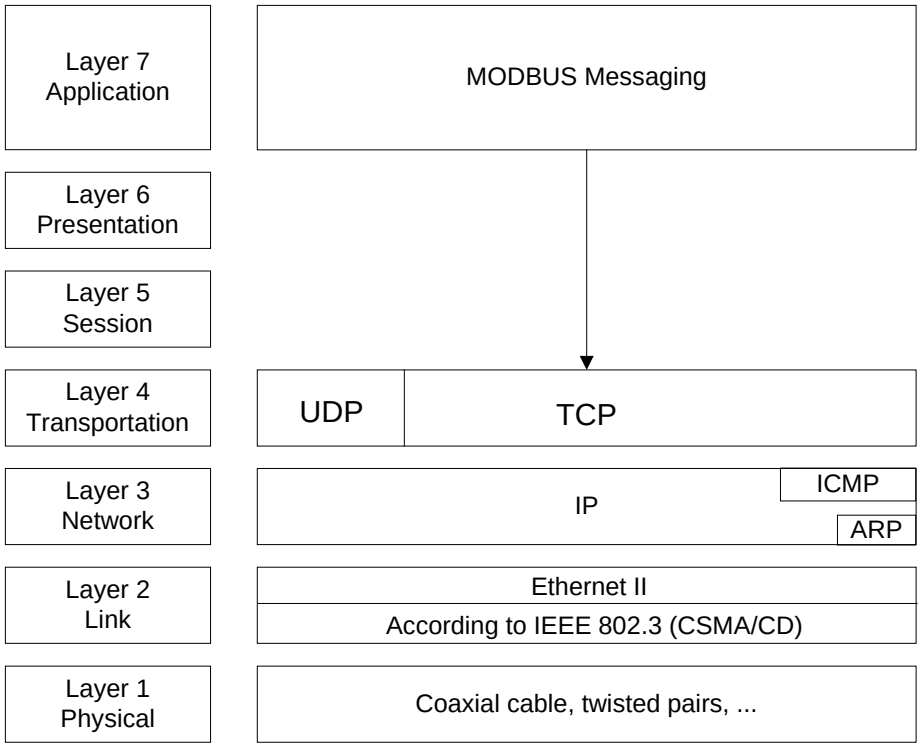
- OSI Model
- Functionality
- Client Multi-Request Management
- Request Processing

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1.1 OSI Model

The MODBUS on TCP/IP protocol uses MODBUS messaging for the application layer and TCP/IP for communication on Ethernet. All layers produced for MODBUS on TCP can therefore be represented by the following OSI model:

Figure 1: TCP/IP Functionality and the OSI Model



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Specifications Used by the Backplane

All TCP/IP protocols are created according to the RFCs (Requests for Comments).
The following table specifies the RFC used.

Table 1: Specifications Used by the Backplane

Layer Name	TCP/IP
Layer 4	TCP : RFC 793 UDP : RFC 768
Layer 3	ICMP : RFC 792 IP : RFC 791 ARP : RFC 826
Layer 2 b	Ethernet II
Layer 2 a	IEEE 802.3 CSMA/CD

Remark: Layers 5 and 6 are not used.

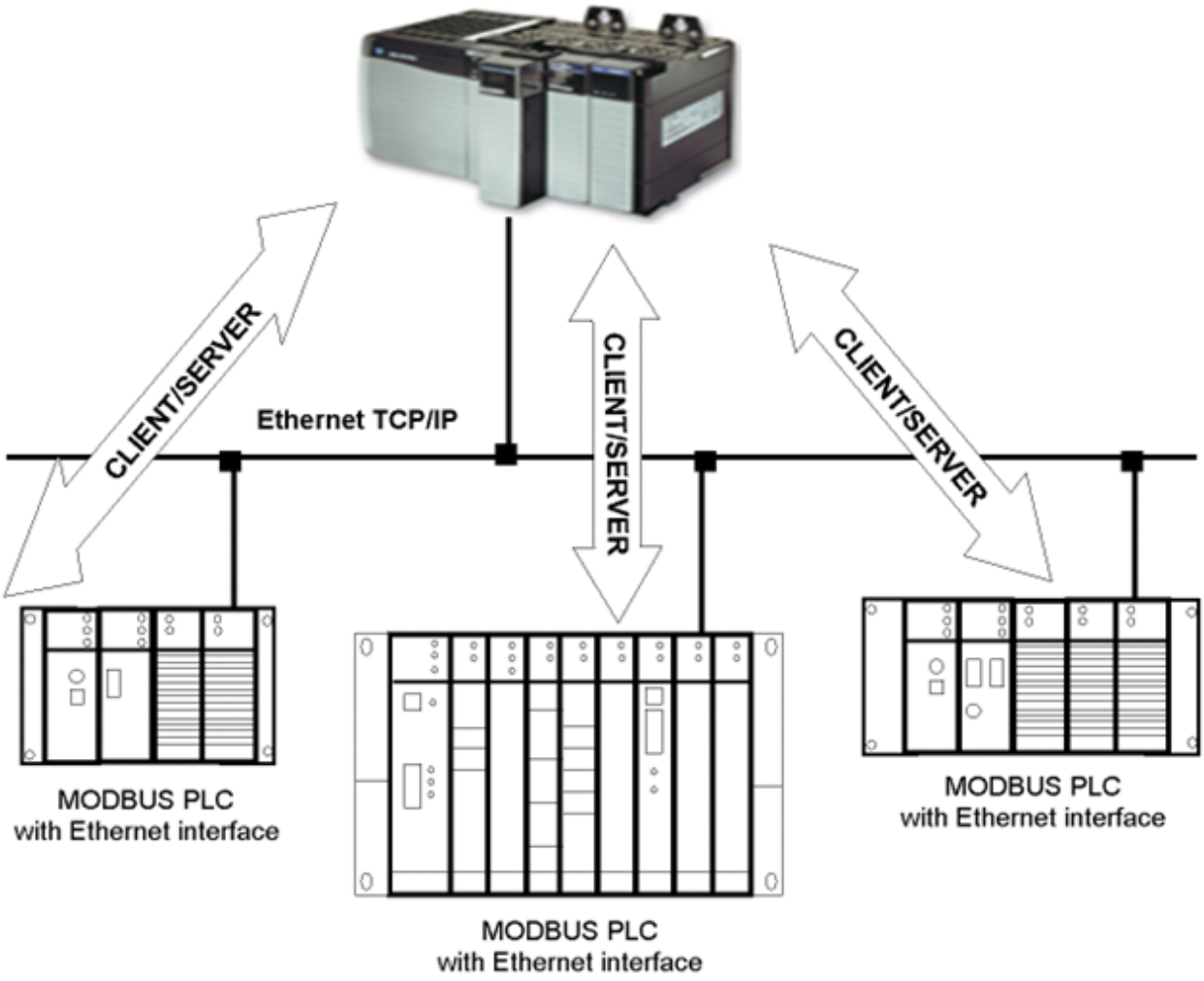
MODBUS messaging (Layer 7) is carried out according to Modicon specifications, which are available on the Modicon website: http://www.modicon.com/openmbus.

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1.2 Functionality

The TCP/IP solution provides the following functionality:

Figure 2: Client/Server Functionality



TCP/IP Functional Modes

- **Multi-Request Client Mode:** used with Ethernet TCP/IP to read and write variables in the PLC's various memory areas. Supports up to 128 Server connections.
- **Server Mode:** used with MODBUS client PLCs to access the database and monitor equipment. Supports up to 60 client connections.

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Note

If using Client and Server mode at the same time then the number of available client connections depends on the number of server connections configured for client. For example, in client mode, 100 server connections have been configured. Then 28 client connections would be available for server mode.

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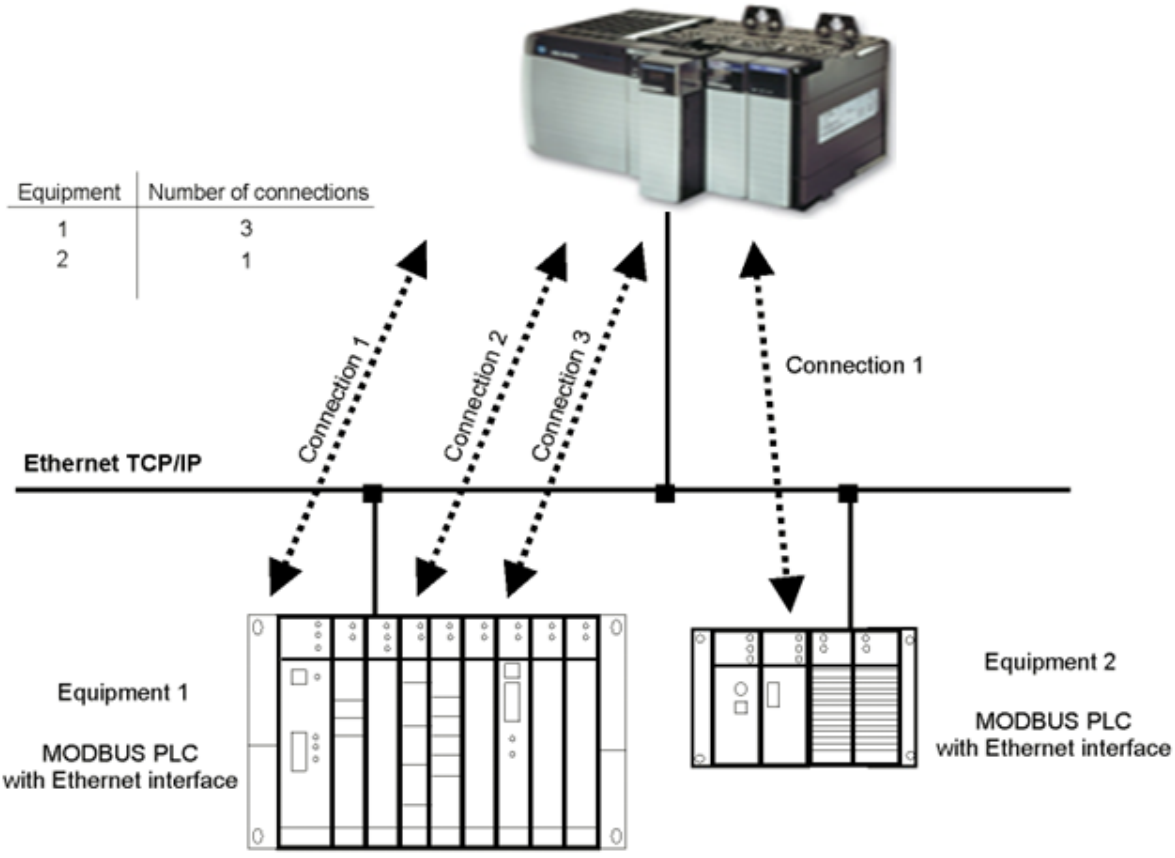
1.3 Client Multi-Request Management

Client Mode can be used to send multiple simultaneous requests to a device before receiving the first response. This operation is called multi-request.

Requests are sent on a virtual communication channel. Each virtual channel corresponds to a TCP connection to a device.

To send multiple requests simultaneously, several connections are used.

Figure 3: Multiple Connections and Multi-Request



Connections are created as required, and existing connections are reused whenever possible.

For example, a request is sent to a device for the first time. Connection No.1 is created. Sometime later, 2 requests must be sent to the device. Connection No.1 is reused and Connection No. 2 is created.

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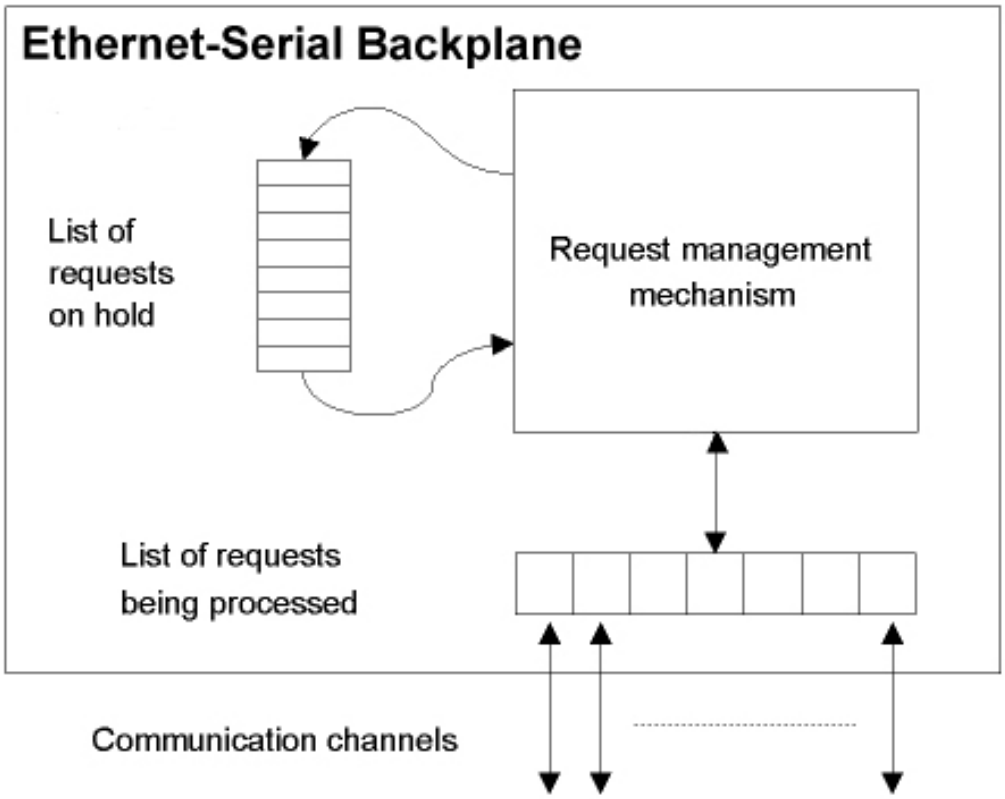
The Backplane manages 128 connections, 30 of which can be used simultaneously for all devices.

The maximum number of simultaneous connections is set in each device's configuration. For devices that don't accept multiple connections, this number is set at 1 and cannot be modified.

A connection remains open as long as data is transmitting or a connection servicing process is active.

1.4 Request Processing

Figure 4: Request Processing Model



A "processed" request is one for which a frame has been sent on the network and a response is expected. Each such request uses a virtual communication channel.

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A request is temporarily placed on hold when no more virtual channels are available. When the channels are released, the on-hold list is progressively emptied.

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2

MODBUS on TCP/IP Functionality

Chapter Sections:

- MODBUS Client to a TSX QUANTUM Server
- MODBUS Client to a TSX PREMIUM Server
- MODBUS Client to a GENERIC MODBUS Server
- MODBUS Server

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2.1 MODBUS Client to a TSX QUANTUM Server

2.1.1 TSX QUANTUM PLC/Ethernet Coupler Reference

TSX QUANTUM processor (Minimum version V2.0):

- Ref.: 140 CPU 113 02
- Ref.: 140 CPU 113 03
- Ref.: 140 CPU 213 04
- Ref.: 140 CPU 424 02

Ethernet driver:

- Ref.: 140 NOE 211 (10 BaseT connection / RJ45)
- Ref.: 140 NOE 251 (fiber optic connection)
- Ref.: 140 NOE 771 (10 BaseT connection / RJ45)

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2.1.2 TSX QUANTUM Addressing Mode

The functions managed by a TSX QUANTUM server device provide access to the variables defined in the following table.

The address of the variable must be converted according to the “Addresses” column below.



Note

There is a difference of 1 between the TSX QUANTUM variable addresses and the configured addresses.

Table 2: TSX QUANTUM Addressing Mode Values

MODBUS Function Code	PLC Variable	Addresses
1	000001 - 065536	0 - 65535
15	000001 - 065536	0 - 65535
2	100001 - 165536	0 - 65535
3	400001 - 465536 400001 - 465536	0 - 65535
	D400001 - D465535 F400001 - F465535	0 - 65534
16	400001 - 465536 400001 - 465536	0 - 65535
	D400001 - D465535 F400001 - F465535	0 - 65534
4	300001 - 365536	0 - 65535

2.1.3 TSX QUANTUM Variables per Exchange

Table 3: TSX QUANTUM Variables per Exchange

Object	Maximum Number	
	Read	Write
Bit	2000	800
Word	125	100
Double/floating word	62	50

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2.2 MODBUS Client to a TSX PREMIUM Server

2.2.1 TSX PREMIUM PLC/Ethernet Driver Reference

TSX PREMIUM processor (Minimum version V3.3):

- Ref.: TSX P57 1xx / PMX P57 1xx
- Ref.: TSX P57 2xx / PMX P57 2xx
- Ref.: TSX P57 4xx / PMX P57 4xx



Note

TSX P57 10 and TSX P57 20 CPUs do not support the Ethernet driver.

Ethernet driver:

- Ref.: TSX ETY 110 (Minimum version V2.6)
- Ref.: TSX ETY 410
- Ref.: TSX ETY 510

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2.2.2 TSX PREMIUM Addressing Mode

The functions managed by a TSX PREMIUM server device provide access to the variables defined in the following table.

Table 4: TSX PREMIUM Addressing Mode Values

Function Code	PLC Variable	Use in the Backplane	
		Addresses	Exchange Type
1	%MX0 - %MX65535	0 - 65535	Read bit
15	%MX0 - %MX65535	0 - 65535	Write bit
3	%MW0 - %MW65535	0 - 65535	Read word
	%MW0 - %MW65535		Read BCD word
	%DW0 - %DW65534	0 - 65534	Read double word
	%FW0 - %FW65534		Read floating word
16	%MW0 - %MW65535	0 - 65535	Write word
	%MW0 - %MW65535		Write BCD word
	%DW0 - %DW65534	0 - 65534	Write double word
	%FW0 - %FW65534		Write floating word

2.2.3 TSX PREMIUM Variables per Exchange

Table 5: TSX PREMIUM Variables per Exchange

Object	Maximum Number	
	Read	Write
Bit	1016	1984
Word	127	124
Double/floating word	63	62

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2.3 MODBUS Client to a GENERIC MODBUS Server

MODBUS messaging on TCP/IP is carried out according to Modicon specifications, which are available on the Modicon website: http://www.modicon.com/openmbus.

However, configuring a generic MODBUS server device enables:

- The definition of the TCP connection port number
- The definition of the maximum number of variables per request (per the MODBUS specifications)
- Preprocessing of double word/floating variables in Intel or Motorola format

This option can be selected when the targeted MODBUS server is a gateway.

2.3.1 GENERIC MODBUS Addressing Mode

The functions managed by a generic MODBUS server enable access to the variables defined in the following table.

Table 6: GENERIC MODBUS Addressing Mode

MODBUS Function Ccode	Item	Use in the Backplane	
		Addresses	Exchange Type
1	B0 - B65535	0 - 65535	Read bit
15	B0 - B65535	0 - 65535	Write bit
2	BI0 - BI65535	0 - 65535	Read input bit
3	W0 - W65535 W0 - W65535	0 - 65535	Read word Read BCD word
	D0 - D65534 F0 - F65534	0 - 65534	Read double word Read floating word
16	W0 - W65535 W0 - W65535	0 - 65535	Write word Write BCD word
	D0 - D65534 F0 - F65534	0 - 65534	Write double word Write floating word
4	WI0 - WI65535	0 - 65535	Read input word

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2.3.2 GENERIC MODBUS Variables per Exchange

Table 7: GENERIC MODBUS Variables Per Exchange

Object	Maximum Number	
	Read	Write
Bit	2040	2040
Word	127	127
Double/floating word	63	63

2.4 MODBUS Server

On the Backplane, the MODBUS server makes a database of 32k words and 32k bits available to MODBUS clients.

MODBUS clients can access the database directly by targeting port number 502 (TCP) of the Backplane. The access mode is identical to that used for a TSX Quantum.

Table 8: MODBUS Server Memory Map Model

Supported Requests	Data Type in the Database	Addresses in the Database	Corresponding MODBUS Variables
Read/write bit	Bit area	0 – 32767 (bit database)	000001 - 032768
Read/write word	Word area	0 – 30207 (word database)	400001 - 430208



Note

Within the 32k word database, offset from 0 to 30207 are reserved for I/O and Status, offset from 30720 to 31743 are reserved for cyclic function Command bit, offset from 31744 to 32767 are reserved for cyclic function State Word.

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Maximum Number of Variables per Exchange

2040 bits
127 words

The database's MODBUS Server functionality can be used to optimize data feedback. Rather than continuously monitoring status-changing variables, devices can use the database to deposit feedback on status changes only (alarm feedback). This operating mode results in the following:

- PLC processors are less solicited
- The network architecture is not as busy
- The data feedback time is minimized

This principle can be used on the MODBUS Server to ensure that the Backplane doesn't work "blindly". For example, if the transmitting device is disconnected, the variables in the database will not change. To avoid this problem, do the following:

1. Using the channel configuration utility, define an address for the Access Status Word in the database. This word will be used to notify the Ladder Logic application when a transmitting device has been disconnected.

2. In the configurator, define a maximum time interval between each time the client device accesses the Backplane's MODBUS Server. When this interval has elapsed, the Access Status Word will announce the disconnection.

3. The application is notified each time a device accesses the Backplane's MODBUS Server in write mode via an "Access Indicator Word". A second device can view the variables for this device and can reset the word to zero so that it will be notified of the next access (or even to tell the transmitting device that this access has been recognized).

For detailed on how to define Access Status Word and Access Indicator Word, please refer to section 4.2

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3

The Console

Chapter Sections:

- Using the Console

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3.1 Using the Console

For instructions on using the Console, refer to the 715-0097___SST-ESR2-CLX-RLL and SST-SR4-CLX-RLL User Reference Guide.

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4

Configuring the MODBUS

Chapter Sections:

- Database Configuration
- Ethernet Connection Overview
- Ethernet Connection Properties
- TCP/IP Protocol Properties
- Equipment Properties
- TSX QUANTUM Server Equipment Properties
- TSX PREMIUM Server Equipment Properties
- GENERIC MODBUS Server Equipment Properties
- MODBUS Client Equipment Properties

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4.1 Database Configuration

The backplane database configuration indicates the specified database offsets, data size and status for updating the module's input, output and status tables with the CLX.

Open the Database Configuration dialog by highlighting Database and selecting Properties command.

It is a word database.

Figure 5: Database Configuration

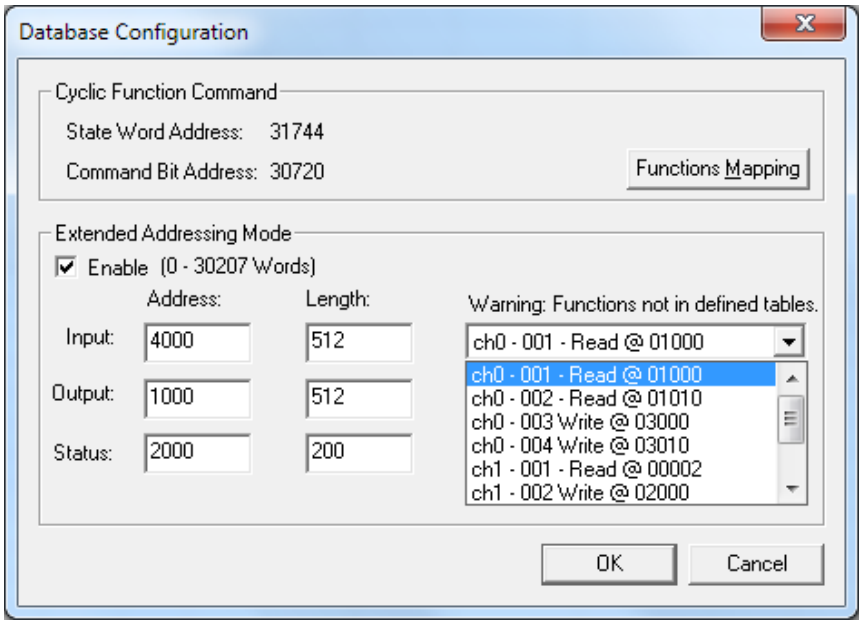


Table 9: Database Configuration Properties

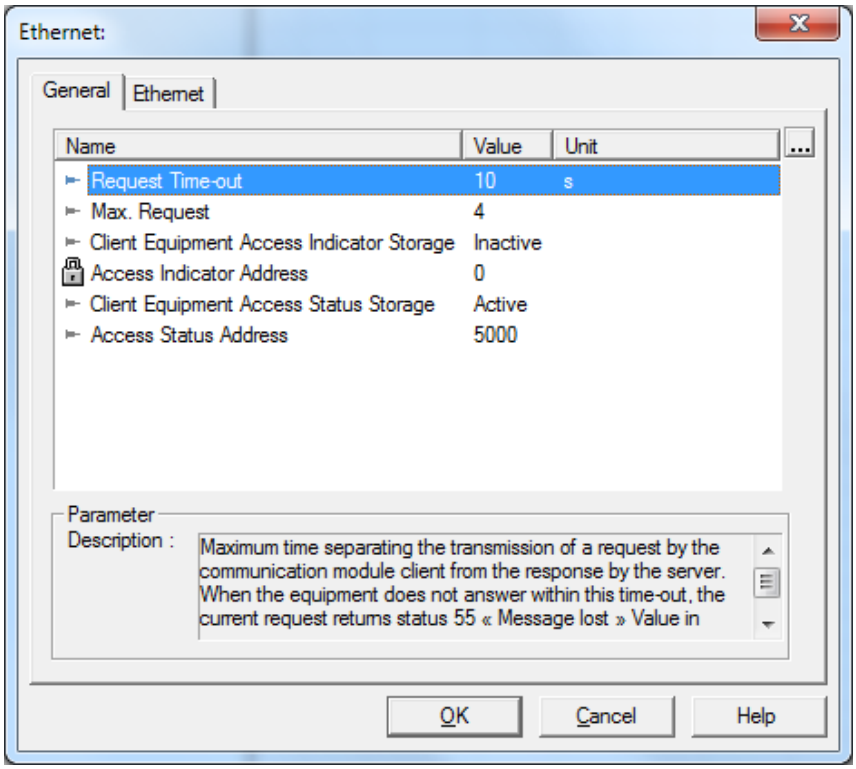
Property	Description
Enable	If it is unchecked, the default address mode is enabled. The input, output and Status fields are grey out with the default settings. If it is checked, the extended address mode is enabled. The input, output and Status fields are configurable.
Input	For default address mode, the input address is 2, length is 248. For extended address mode, the input address can be ranged from 0 to 29695, the length must be at least 512.
Output	For default address mode, the output address is 250, length is 248. For extended address mode, the output address can be ranged from 0 to 29695, the length must be at least 512.
Status	For default address mode, the status address is 500, length is 200. For extended address mode, the input address can be ranged from 0 to 29695, the length can be from 0 to 1024.

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Property	Description
Functions not in defined tables	It displays the configured cyclic functions that are not in defined input and output tables. These cyclic functions can be only communicating with CIP messages in CLX.
Functions Mapping	Click on this button, it will open a text file which lists all configured cyclic functions' state word address and command bit address. For detail, refer to section 7.3.7 in 715-0097___ SST-ESR2-CLX-RLL and SST-SR4-CLX-RLL User Reference Guide.

4.2 Ethernet Connection Overview

Figure 6: Ethernet Connection Dialog, General Tab



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Table 10: Ethernet Connection Dialog, General Tab Properties

Property	Description/Meaning
Request Time out	This is the maximum time between the sending of a function by the client and the response from the server. If the equipment does not answer within this period, the current request returns status 55, “Message lost” Value in seconds, from 2 to 255; default value 10.
Maximum Request	Sets the maximum number of requests to be processed. Requests exceeding this number are placed on a waiting list until a request being processed is released. The number of requests being processed indicates how many requests are being sent simultaneously on the network. These requests contribute to the network load. By modifying the maximum number of requests to be processed, this load can be regulated. One of two possible cases may then arise for applications, including a Backplane and PLCs : • Dedicated network: the default maximum optimizes the performance of the communication support. • Network shared with workstations: in case of network overload (>30%), the maximum number of requests pending must be reduced. This is done to preserve the passband for the other stations. Value from 1 to 32; default value 4.
Access Status Address	Activates the "Client device access indicator storage" field to validate operation and specify the storage address of the databases 60 access status words. Value from 0 to 30207; default value 0. This option is used to monitor the client equipment’s access interval to the Backplane’s server mode. If the “Access monitoring time-out” configured in the equipment (see configuration of client equipment) is exceeded, the Backplane indicates its absence in the word corresponding to the client equipment number. 1st word ⇨ Client equipment number 0 … 60th word ⇨ Client equipment number 59 Value of the access status word : 0: the equipment accesses the server in the defined time, 4: the client equipment accesses inaccessible data in the server, 33: the client equipment does not access the server within the defined time, 36: the targeted equipment is not configured

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Property	Description/Meaning
Access Indicator Address	Activates the "Client device access status storage" field to validate operation and specify the storage address of the databases 60 access indicator words. Value from 0 to 30207; default value 0. Whenever a client has write access on the Backplane's slave, the word corresponding to the number of client equipment is incremented. 1st word ⇨ Client equipment number 0 .. 60th word ⇨ Client equipment number 59 This word reaches a maximum at 65535. You must reset it to zero so that it can begin tabulating access again.



Note

The Access Indicator Address and Access Status Address should not interfere with word register address. Each of the Access Indicator Address and Access Status Address reserves 60 words.

In default address mode, if the Access Status Address is 0, it would interfere with 400001- 400060.

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4.3 Ethernet Connection Properties

Figure 7: Ethernet Connection Dialog, Ethernet Tab

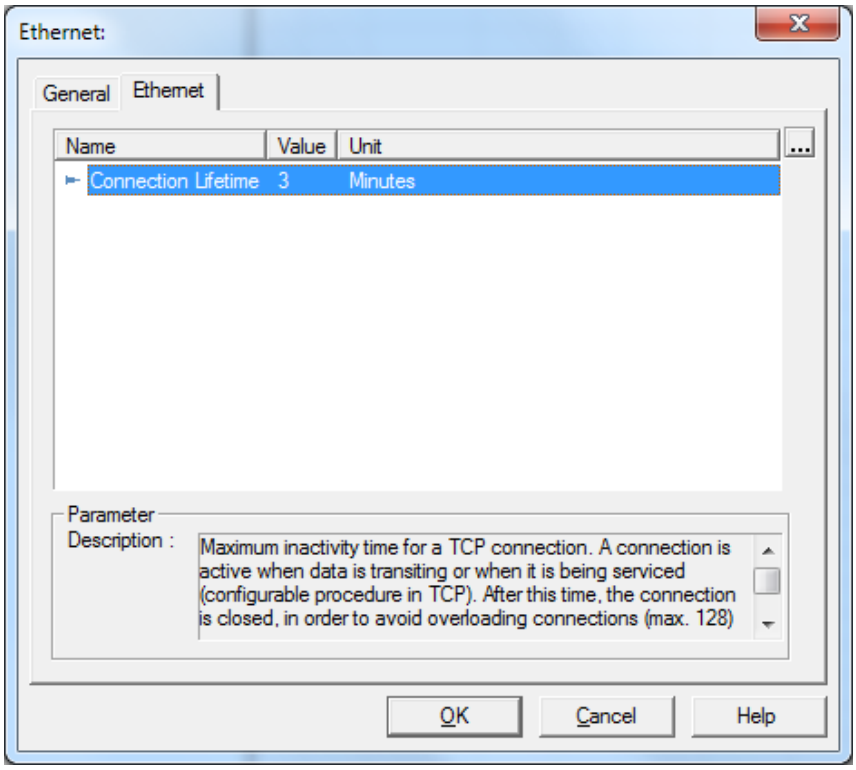


Table 11: Ethernet Connection Dialog, Ethernet Tab Properties

Property	Description/Meaning
Connection Lifetime	Maximum inactivity time for a TCP connection. A connection is active when data is transmitting or when it is being serviced (configurable procedure in TCP). After this time, the connection is closed to avoid overloading (max. 128) when the partner is absent. Value in minutes, from 2 to 60; default value 3.

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4.4 TCP/IP Protocol Properties

Figure 8: TCP/IP Advanced Parameters

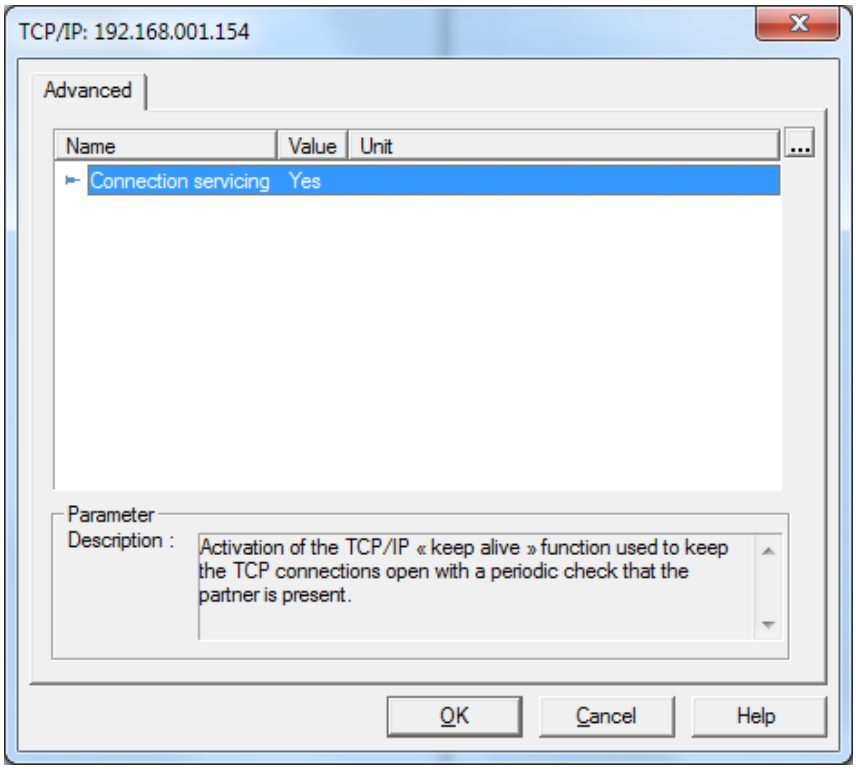


Table 12: TCP/IP Protocol Properties

Property	Description/Meaning
Connection Servicing	Activation of the TCP/IP's "keep alive" function used to keep the TCP connections open with a periodic check that the partner is present.

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4.5 Equipment Properties

Generally, all server and client devices consist of a common “Equipment Designation”, a “Network Properties” zone, and a “Messaging Parameters” area.

4.5.1 Equipment Designation Zone

Figure 9: Equipment Designation Zone

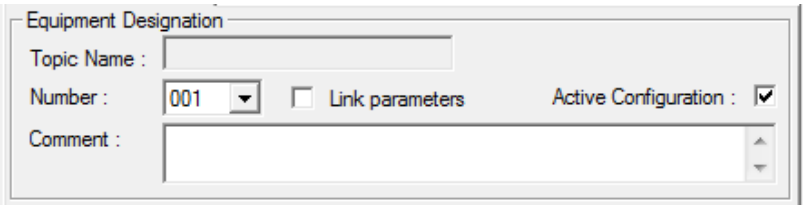


Table 13: Equipment Designation Zone Properties

Property	Description/Meaning
Equipment Number	Equipment number to be chosen from the list of non-configured equipment. Value: • 000 to 127 for server equipment • 000 to 59 for client equipment
Link Parameters	Tick this box to activate the link between the device number and the device's IP address. The device number will then correspond to the IP address’s station number. Depending on the device type, this number may be linked to another messaging parameter.
Active Configuration	Checkbox to activate the interface’s equipment configuration. Used to delete an equipment from the Backplane while keeping its configuration in the Console.
Comment	Free text about an equipment. The maximum number of characters is 80.

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4.5.2 Network Properties Zone

Figure 10: Network Properties Zone

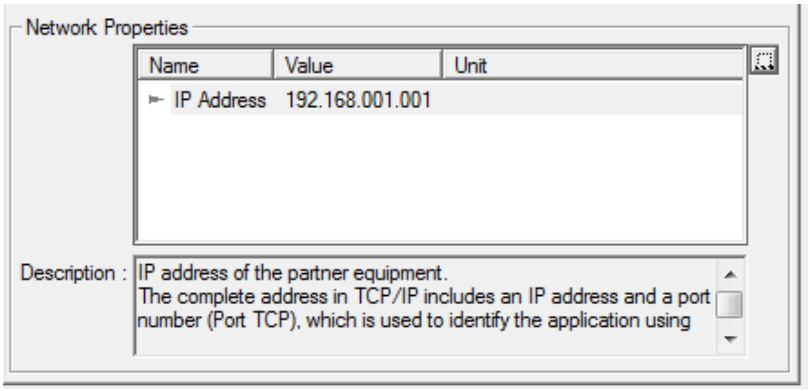


Table 14: Network Properties

Property	Description/Meaning
IP Address of the Equipment	IP address of the partner equipment. The complete address in TCP/IP includes an IP address and a port number (Port TCP), which is used to identify the application using the connection. Generally, the port number is predefined by the PLC manufacturer. However, certain types of equipment allow this port to be modified. The relevant equipment has an "Equipment TCP port" entry zone in the Network parameters box.

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4.6 TSX QUANTUM Server Equipment Properties

4.6.1 TSX QUANTUM General Server Configuration

After selecting the TSX QUANTUM equipment, the equipment number and the network parameters, you must define the “Messaging Parameters Zone”.

Figure 11: TSX QUANTUM General Configuration Dialog

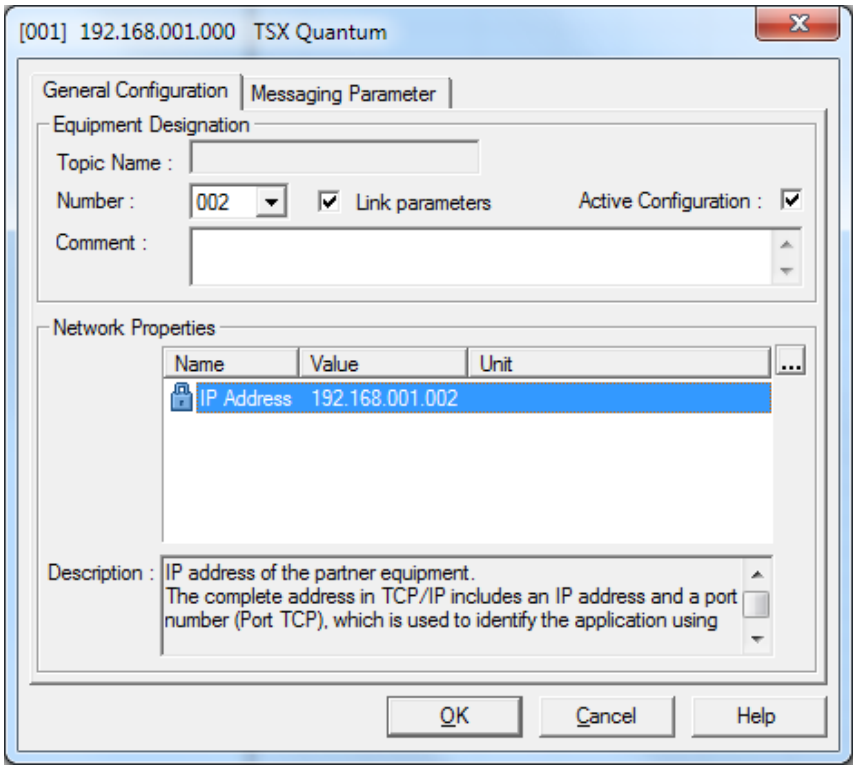


Table 15: TSX QUANTUM, General Configuration Properties

Property	Description/Meaning
Equipment Description	Refer to Section 4.4, Equipment Properties.
Network Properties	Refer to Section 4.4.2, Network Properties Zone.

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4.6.2 Messaging Parameters

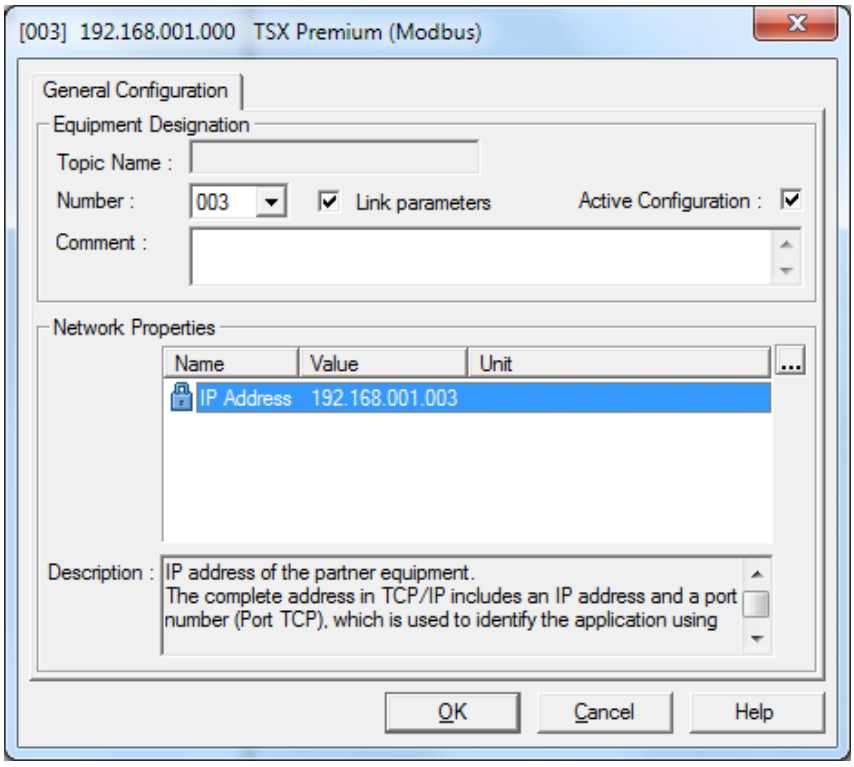
Table 16: Messaging Parameters Properties

Property	Description/Meaning
Maximum Number of Simultaneous Requests	This parameter allows you to adjust the maximum number of simultaneous requests sent by the Backplane's client to an server equipment. Value from 1 to 4; default value 1.

4.7 TSX PREMIUM Server Equipment Properties

After selecting the TSX PREMIUM equipment, the equipment number and the network parameters, you must define the "Messaging Parameters" zone:

Figure 12: TSX PREMIUM General Configuration Tab



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Table 17: Messaging Parameters Properties

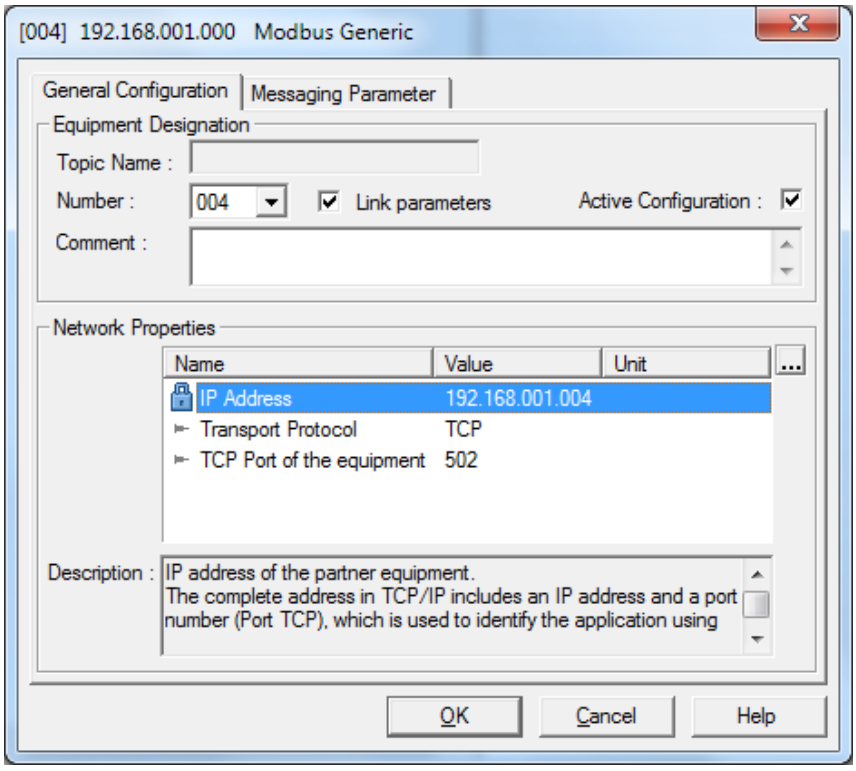
Property	Description/Meaning
Equipment Description	Refer to Section 4.4, Equipment Properties.
Network Properties	Refer to Section 4.4.2, Network Properties Zone.

4.8 GENERIC MODBUS Server Equipment Properties

4.8.1 Generic MODBUS General Server Configuration

After selecting the GENERIC MODBUS equipment, the equipment number and the network parameters, you must define the “Messaging Parameters” zone:

Figure 13: Generic MOBUS General Server Configuration



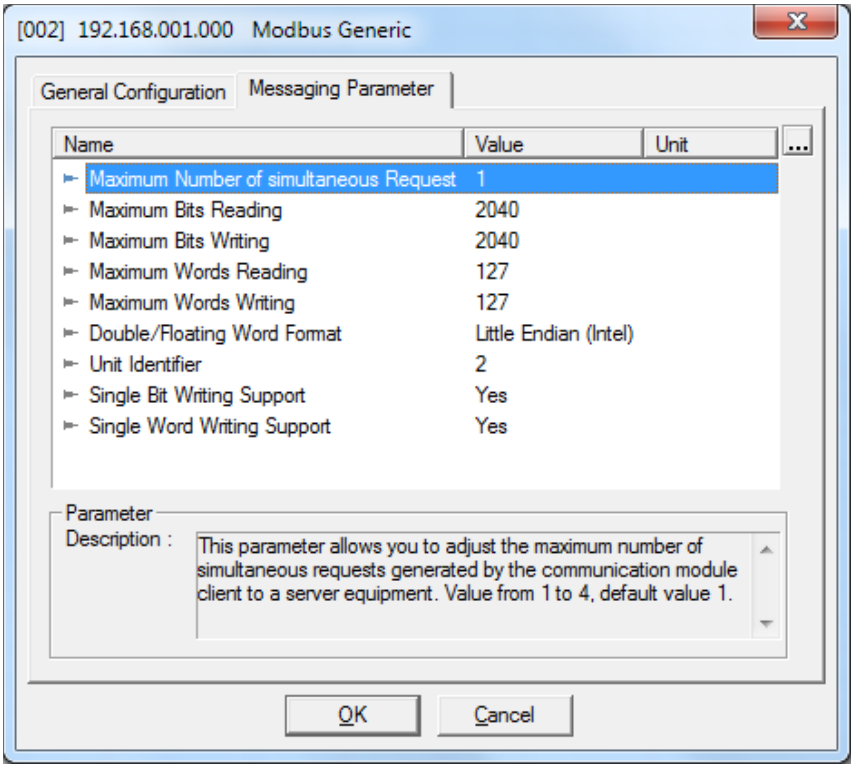
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Table 18: Generic MODBUS General Server Configuration Properties

Property	Description/Meaning
Equipment Description	Refer to Section 4.4, Equipment Properties.
Network Properties	* IP address of the equipment Refer to Section 4.4.2, Network Properties Zone. * Transport protocol Choose UDP or TCP protocol for your network. * TCP port of the equipment. Defines the TCP port number of the remote equipment. The default value (502) corresponds to that in the MODBUS specifications. Value from 1 to 65535; default value 502.

4.8.2 GENERIC MODBUS Messaging Parameters

Figure 14: GENERIC MODBUS Messaging Parameters



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Table 19: GENERIC MODBUS Messaging Parameters Properties

Property	Description/Meaning
Maximum Number of Simultaneous Requests	This parameter allows you to adjust the maximum number of simultaneous requests sent by the Backplane's client to an server equipment. Value from 1 to 4; default value 1.
Double/Floating Word Format	Defines the format used for these variables in the PLC.
Unit Identifier	On TCP/IP, the MODBUS "slave address" field is replaced by a single-byte "Unit Identifier", which may be used to communicate via devices that use a single IP address to support multiple independent end units, such as bridges and gateways.

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4.9 MODBUS Client Equipment Properties

4.9.1 MODBUS Client Equipment General Configuration

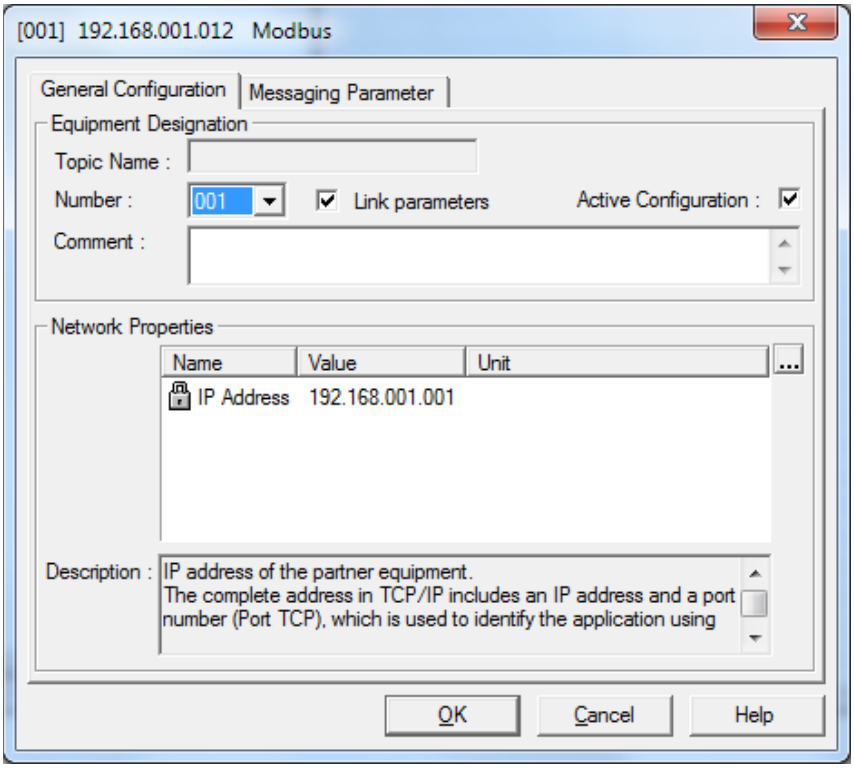
Note

Access to the Backplane's server functionality requires no configuration. The client targets the Backplane's IP address on port TCP number 502 (access identical to that for a PLC).

This configuration is only required to monitor clients of the Backplane server (access time-out and writing in the database). For details, Refer to Section 4.7, GENERIC MODBUS Server Equipment Properties.

After selecting the equipment number and network parameters, you must define the "Messaging Parameters" zone.

Figure 15: MODBUS Client Equipment General Configuration Tab



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Table 20: MODBUS Client Equipment Properties

Property	Description/Meaning
Equipment Description	Refer to Section 4.4, Equipment Properties.
Network Properties	Refer to Section 4.4.2, Network Properties Zone.

4.9.2 MODBUS Client Equipment Messaging Parameters

Table 21: MODBUS Client Equipment Messaging Properties

Property	Description/Meaning
Equipment Description	Definition of the access time-out to the Backplane's MODBUS server. Value from 0 to 65535, default value 5. This value determines the maximum time interval between the client's accesses to the Backplane in server mode. After this delay, its absence will be automatically indicated in the database's Access status words.

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5

Example of a MODBUS TCP Equipment Implementation

Chapter Sections:

- Description
- Configuring the Backplane
- Configuring the Equipment
- Using the MODBUS Equipment

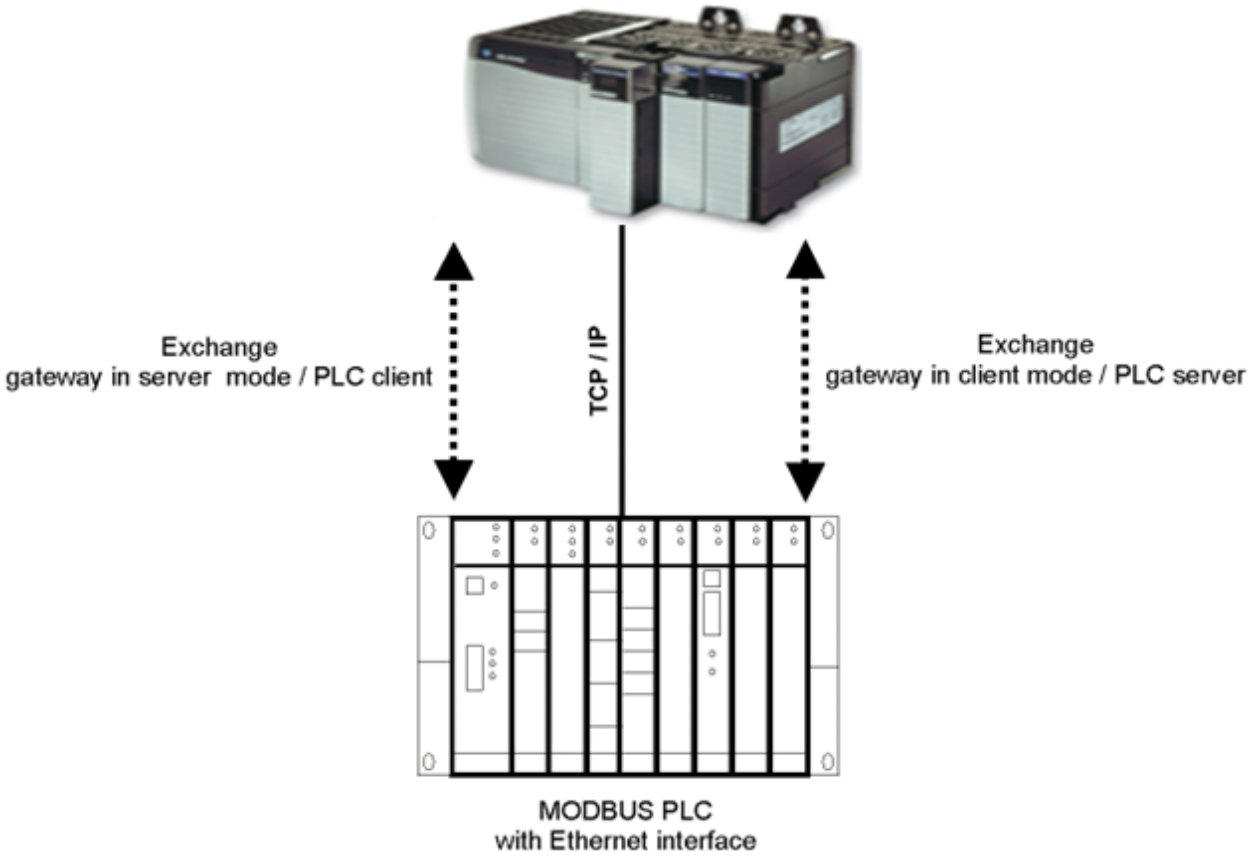
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5.1 Description

In this example, 2 types of exchange are configured:

- The Backplane is a MODBUS PLC client and can read or write words in the PLC
- The Backplane is a MODBUS server and the PLC can read or write words in the database

Figure 16: Backplane as MODBUS PLC Client and MODBUS Server



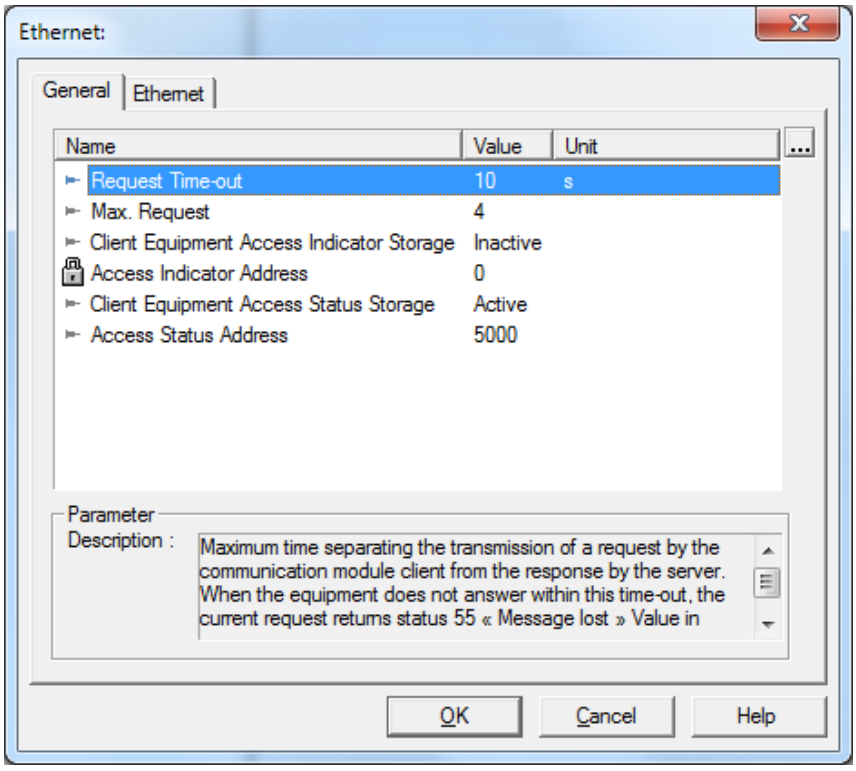
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5.2 Configuring the Backplane

To configure the Backplane, follow these steps:

1. Run the Configuration Console.
2. Click on Channel 0 to open the User Configuration area.
3. Double-click on “Ethernet”. The Ethernet Properties Dialog box is displayed.

Figure 17: Ethernet Properties Dialog

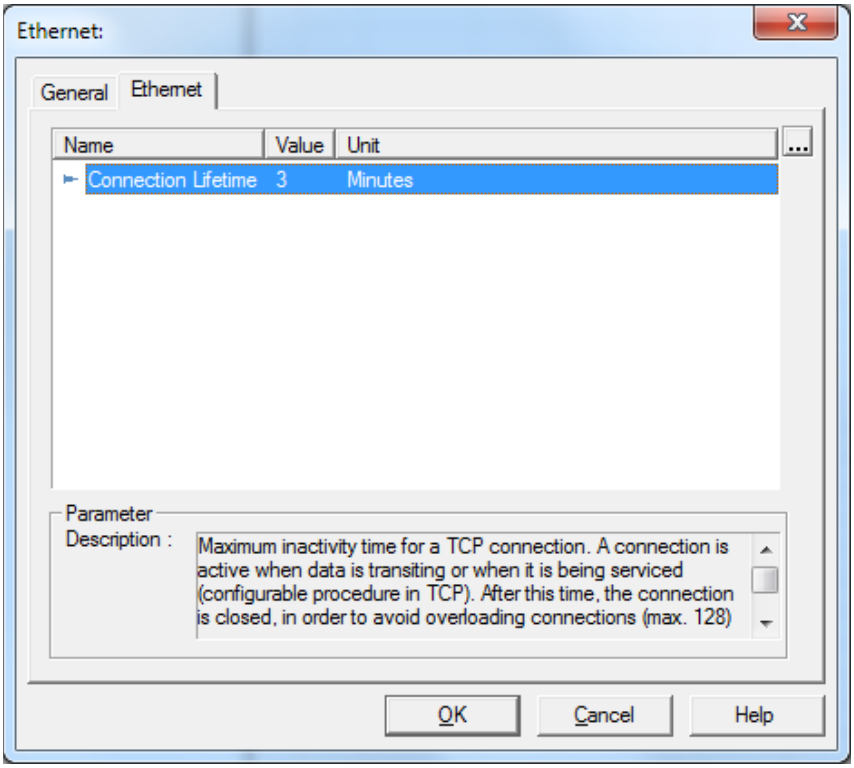


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4. To monitor the Client connection, select “Active” for both “Client Equipment Access” fields and set the status words’ addresses in the database.

5. In the Ethernet parameters, configure the connection type.

Figure 18: Ethernet Parameters Tab



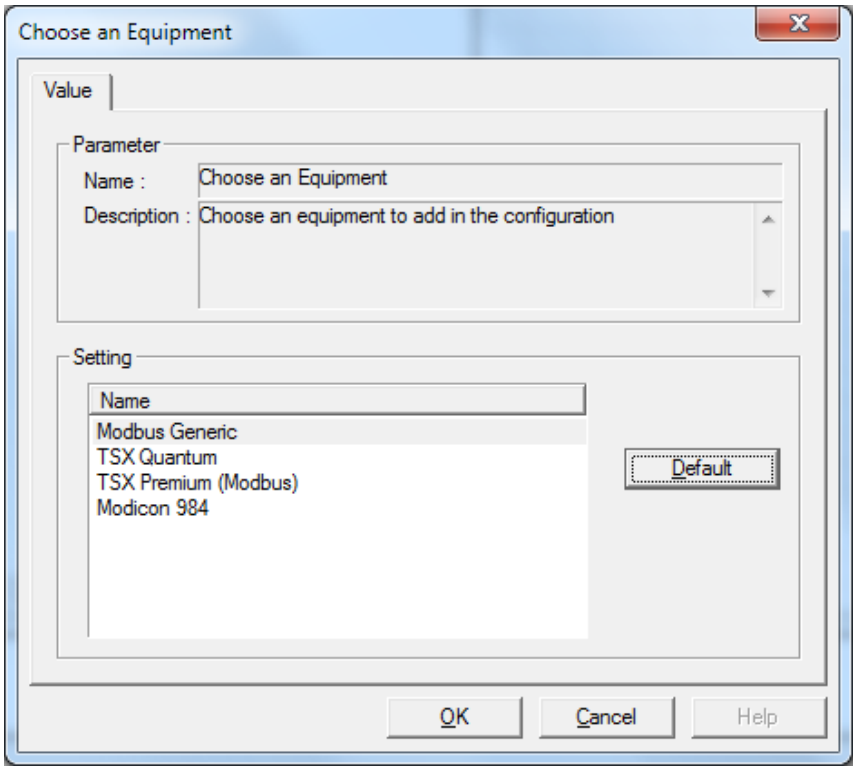
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5.3 Configuring the Equipment

To configure the devices, follow these steps:

1. Right-click on Server Equipments and select Insert. The Choose an Equipment dialog is displayed.

Figure 19: Choose an Equipment Dialog

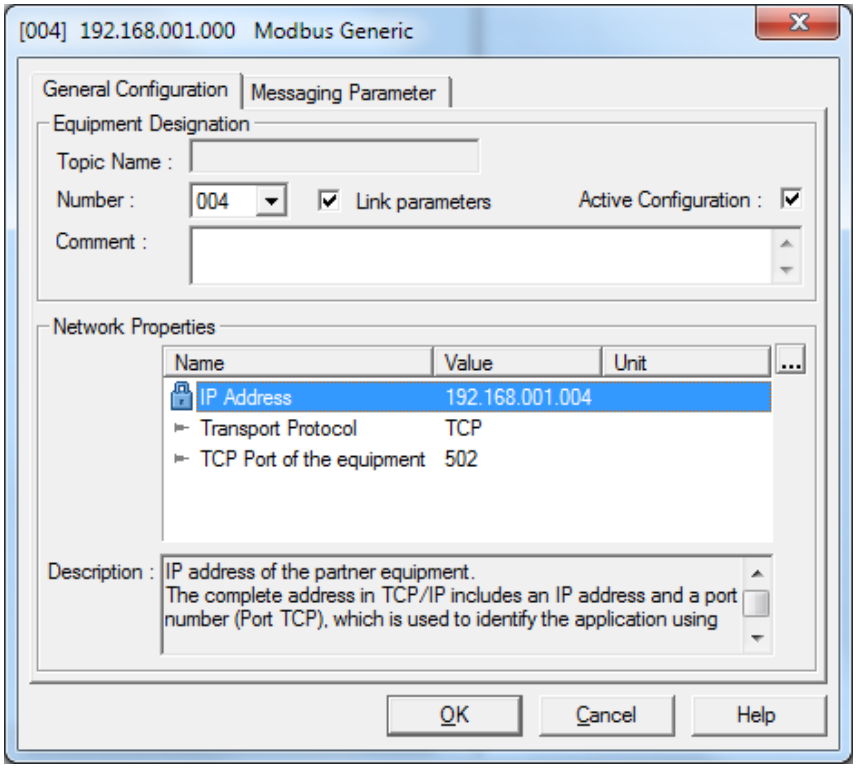


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2. Select MODBUS Generic. This equipment represents the PLC seen by the Backplane in Client mode.

3. Click OK. The server Equipment dialog is displayed.

Figure 20: MODBUS Generic Server Equipment Dialog



Note

In this example, the messaging parameters are not modified.

4. Configure the Server Equipment and click OK.

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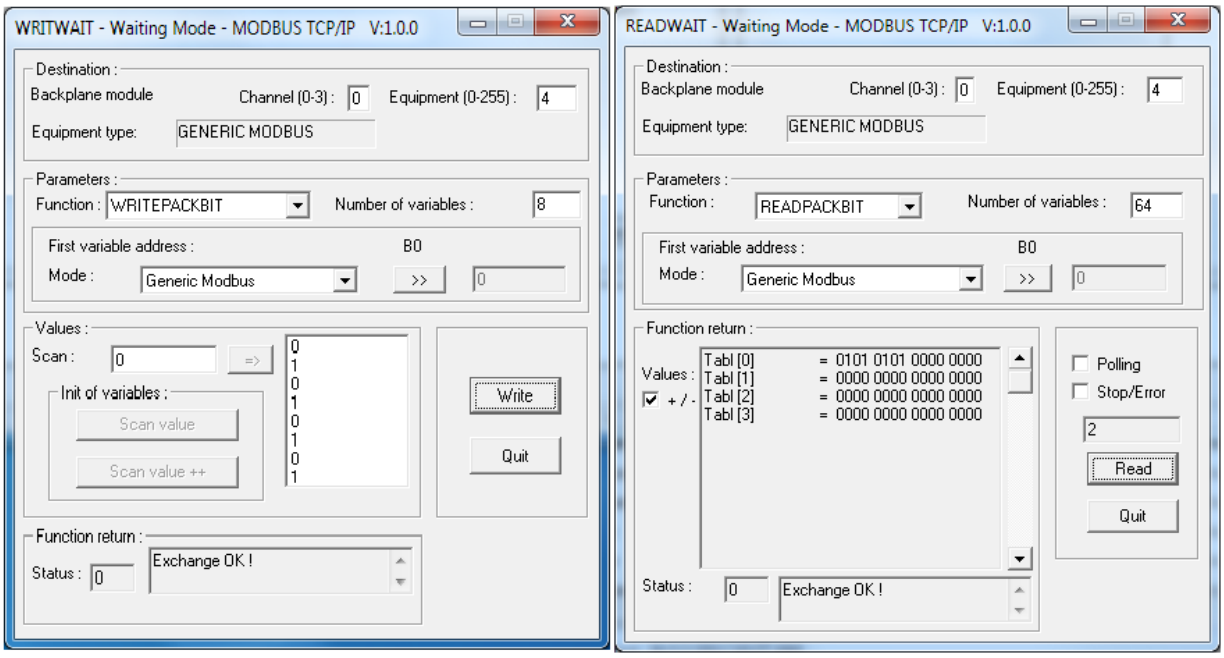
5.4 Using the MODBUS Equipment

To use the MODBUS equipment, follow these steps:

1. From the Configuration Console, click on  to run the module initialization. The new configuration is loaded.

2. To access the PLC, run the READWAIT and WRITWAIT tools.

Figure 21: READWAIT and WRITWAIT Tools





Note

If the PLC is configured to access the Backplane automatically, you can monitor the database's check words operation via the GETDB utility.

For information on READWAIT and WRITWAIT functions, see Section 6.1, READWAIT and WRITWAIT Functions.

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6

Master Channel Functions

Chapter Sections

- READWAIT and WRITWAIT Functions
- Cyclic Functions

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6.1 READWAIT and WRITWAIT Functions

These functions can be called through READWAIT and WRITWAIT tools.

Table 22: Supported Functions

Read Access	Write Access	Data Type
readpackbit	writepackbit	Internal bit
readpackibit		Input bit
readword	writeword	Word
readiword		Input Word
readwordbcd	writewordbcd	Word in BCD format
readdword	writedword	Double word
readfword	writefword	Float

6.2 Cyclic Functions

These functions are used while configuring cyclic functions in the Console.

Table 23: Cyclic Function Type

Variable in the Device	Function Type	
	Read	Write
Packed bits	X	X
Packed input bits	X	
Words	X	X
BCD words	X	X
Input words	X	
Floating IEEE 32 bits	X	X

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7

Image Variable Items

Chapter Contents:

- Presentation
- Standard Descriptor
- MODBUS Modicon TSX Descriptor
- Télémécanique TSX/PMX Descriptor
- Generic MODBUS Descriptor

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7.1 Presentation

Image variable items are syntaxes that allow access to the variables. They are used during the configuration of cyclic functions and via the ReadWait and WriteWait tools.

The most suitable default descriptor is automatically determined by the server. With the MODBUS protocol on TCP/IP, the default descriptors are:

- The MODBUS Modicon TSX descriptor for TSX QUANTUM devices
- The Télemécanique TSX/PMX descriptor for TSX PREMIUM devices
- The Generic MODBUS descriptor for GENERIC MODBUS devices

You do, however, have the option of using a difference descriptor, such as the standard one.

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7.2 Standard Descriptor

The standard descriptor can be used to access all devices that do not have specific descriptors. The address field in the "item" name can be up to 10 digits long, enabling a linear address from 0 to 4 gigabytes to be composed.

Table 24: Standard Descriptor

Data Type	Unitary Mode
Internal bits	Bx
Input bits	BIx
Internal words	Wx
Bits in the internal words	Wx.b
Input words	WIx
Internal double words	Dx
Internal floating words	Fx

Remark: the limit values for parameters n and l depend on the protocol.

For information on variable addressing, refer to:

- Section 2.1.2, TSX QUANTUM Addressing Mode
- Section 2.2.2, TSX PREMIUM Addressing Mode
- Section 2.3.1, GENERIC MODBUS Addressing Mode

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7.2.1 Internal Bits => Bx (Type: BIT)

x: Number of the first bit.

Example: B4

7.2.2 Input Bits => BIx (Type: BIT)

x: Number of the first bit.

Example: BI4

7.2.3 Internal Words => Wx (Type: 16-Bit WORD)

x: Number of the first word.

Example: W4

7.2.4 Bits in the Internal Words => Wx.b (Type: BIT)

For this syntax, the maximum number of variables per frame is:

- In read: 128*16
- In write: 1

x: Number of the first word.

b: Rank of the bit in the word (0 to 15)

Example: W4.5

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7.2.5 Input Words => WIx (Type : 16-Bit WORD)

x: Number of the first word.

Example: WI4

7.2.6 Internal Double Wwords => Dx (Type: 32-Bit WORD)

x: Number of the first double word.

Example: D4

7.2.7 Internal Floating Words => Fx (Type: REAL IEEE 32 Bits)

x: Number of the first floating word.

Example: F4

7.3 MODBUS Modicon TSX Descriptor

This descriptor can only be used for access to TSX QUANTUM PLCs.

Table 25: MODBUS Modicon TSX Descriptor

Data Type	Unitary Mode
Internal bits	0xxxx
Input bits	1xxxx
Internal words	4xxxx
Bits in the internal words	4xxxx.b
Input words	3xxxx
Internal double words	D4xxxx
Internal floating words	F4xxxx

Remark: The limiting values of parameters n and 1 depend on the protocol.

For variable addressing, refer to Section 2.1.2, TSX QUANTUM Addressing Mode.

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7.3.1 Internal Bits => 0xxxxx (Type: BIT)

x: Number of the first bit.

Example: 000004

7.3.2 Input Bbits => 1xxxxx (Type: BIT)

x: Number of the first bit.

Example: 100004

7.3.3 Internal Words => 4xxxxx (Type: 16-Bit WORD)

x: Number of the first word.

Example: 400004

7.3.4 Bits in the Internal Words => 4xxxxx.b (Type: BIT)

For this syntax, the maximum number of variables per frame is:

- In read: 128*16
- In write: 1

x: Number of the first word.

b: Rank of the bit in the word (0 to 15).

Example: 400002.5

7.3.5 Input Words => 3xxxxx (Type: 16 Bit WORD)

x: Number of the first word.

Example : 300004

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7.3.6 Internal Double Words => D4xxxxx (Type: 32-Bit WORD)

x: Number of the first double word.

Example: D400002

7.3.7 Internal Floating Words => F4xxxxx (Type: 32-Bit IEEE REAL)

x: Number of the first floating word.

Example: F400002

7.4 Télémécanique TSX/PMX Descriptor

This descriptor can be used only to access TSX PREMIUM, under the MODBUS protocol.

Table 26: Télémécanique TSX/PMX Descriptor

Data Type	Unitary Mode
Internal bits	%MXx
Internal words	%MWx
Bits in the internal words	%MWx.b
Internal double words	%MDx
Internal floating words	%MFx

Remark: the limit values for parameters n and l depend on the protocol. For variable addressing, refer to Section 2.2.2, TSX PREMIUM Addressing Mode.

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7.4.1 Internal Bits => %MXx or %Mx (Type: BIT)

x: Number of the first internal bit.

Example: %MX4 or %M4

7.4.2 Internal Words => %MWx (Type: 16-Bit WORD)

x: Number of the first word (0 to 32767).

Example: %MW4

7.4.3 Bits in the Internal Words => %MWx.b (Type: BIT)

For this syntax, the maximum number of variables per frame is:

- In read: 60*16
- In write: 1

x: Number of the word (0 to 32767)

b: Rank of the bit in the word (0 to 15)

Example: %MW4.5

7.4.4 Internal Double Words => %MDx (Type : 32-Bit WORD)

x: Number of the first double word (0 to 32766).

Example: %MD4

7.4.5 Internal Floating Words => %MFx (Type : 32-Bit IEEE)

x: Number of the first floating word (0 to 32766).

Example: %MF4

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7.5 Generic MODBUS Descriptor

The Generic MODBUS descriptor is identical to the standard descriptor. It ensures optimization by grouping word, double word and floating word variables into the same MODBUS frame.

For details, refer to Section 7.2, Standard Descriptor.

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A

Function Return Statuses

Appendix Sections:

- Function Return Statuses

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A.1 Function Return Statuses

The various functions return a status word to:

- Guarantee the quality of the request
- Diagnose the cause of a failure

The meaning of the status word value is detailed in the following tables. In addition to the general meaning, "Further information" allows you to direct your diagnosis according to the protocol used.

A.1.1 General Statuses

Table 27: General Return Statuses

Word Value	Meaning
0	No anomaly detected. The function has been properly executed.
1	Unknown function. The requested function is not supported.
2	Bad address. The address of the variable that you are soliciting is bad.
3	Bad data. Additional information: Function : BINBCD, BCDBIN. At least one of the values accessed is not in BCD format (0<= value <= 9999).
4	Inaccessible data.
32	Bad parameter passed to the function. Bad number of variables.
45	Communication program not configured. Initialize the Backplane before using it by calling the initialization tool, via the configuration Console.
46	Master/client function directed at a channel configured in slave/server, or vice versa.
47	No Backplane.
51	Driver system problem.
66	Not enough memory.

Remark: the negative statuses are information codes.

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A.1.2 Protocol Statuses

Table 28: Protocol Statuses

Word Value	Meaning
0	No anomaly detected. The function was executed correctly.
2	Bad address. The address of the variable that you are soliciting is incorrect. Additional information: Messaging on TCP/IP Ethernet: Returned by the coupler of the destination equipment. Access to a variable with a bad address or a number exceeding the PLC zone.
3	Bad data. Additional information: Incoherent frame content.
4	Inaccessible data. Additional information: • The physical address does not exist. Non-existent module, or protected data. • The requested data length exceeds the remote device's communication capabilities
6	Slave device Busy. Additional information: Mainly used in conjunction with programming commands. The remote device has no ressources to answer to the request. • Resend the request • Check remote device's the timing / ressources with the manufacturer
7	Negative Acknowledge. Additional information: The received request does not correspond to the MODBUS standard, or the function code is not supported by the remote device. Can also appear if all available Ethernet connections to the devices are used. • Check the quality of the Ethernet communication • Check the MODBUS function code supported by the remote device and the function code sent • Check the Ethernet connections available for this device and the number of connection already used. In the equipment properties, reduce the number of simultaneous requests allowed.

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Word Value	Meaning
10	Backplane path unavailable. Additional information: Specialized use, in conjunction with MODBUS Serial / MODBUS TCP Backplanes. Indicates that the Backplane was unable to allocate a MODBUS TCP PATH for processing the request. This usually means that the Backplane is misconfigured. • Check the MODBUS Serial / MODBUS TCP Backplane configuration • Check the “Unit identifier” parameter in the Configuration Console, under the equipment properties
11	Backplane target device failed to respond. Additional information: Specialized use, in conjunction with MODBUS Serial / MODBUS TCP Backplanes. Indicates that no response was obtained from the target device. This usually means that the device is not present on the network. • Check the MODBUS Serial / MODBUS TCP Backplane configuration • Check the MODBUS Serial / MODBUS TCP Backplane communication link to the remote device • Check the “Unit identifier” parameter in the Configuration Console, under the equipment properties
33	Time-out fault. Additional information: Equipment configured but not connected on the network, wiring problem, CPU not running communication blocks; connection not declared or badly declared in the CPU. Bad equipment or Backplane IP address. Verify that the format of the Ethernet field configured in the remote device is “ETHERNET II”.
36	Equipment not configured. Define the device in the console and restart the Backplane.
49	Queue time-out fault The request could not be sent due to lack of resources (no communication channels available). This period corresponds to 4 times the value of the time-out for the requests being processed. • Increase the value of the “time-out for requests being processed” • Increase the maximum number of simultaneous requests on the targeted device.
55	Response time-out exceeded, message lost. Additional information: Time-out exceeding the value of the “time-out of requests being processed”. Connection established and question acknowledged, but no response.

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Word Value	Meaning
66	Not enough memory. Additional information: Insufficient resources for an additional connection.
70	Connection terminated by the remote device Additional information: • Connection aborted following a connection problem. Wiring problem, CPU in stop, or the CPU does not run communication blocks. • The targeted equipment does not support this messaging.

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B

Warranty and Support

Appendix Sections:

- Warranty
- Technical Support

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B.1 Warranty

For warranty information, refer to
http://www.molex.com/images/woodhead/woodhead_limited_warranty.pdf.

B.2 Technical Support

Please ensure that you have the following information readily available before calling for technical support:

- Software version
- Computer's make, model, CPU speed and hardware configuration (cards installed)
- Operating system type and version
- Details of the problem you are experiencing: firmware module type and version, target network and circumstances that may have caused the problem

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B.2.1 Getting Help

Technical support is available during regular business hours by telephone, fax or email. The Molex web site contents also useful information available by clicking the link below:

Molex Support and Download

- Downloads center
- Support Request Form
- Knowledge Base
- Worldwide technical support contacts

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Glossary

ARP
Address Resolution Protocol.

ASCII
American Standard Code for Information Interchange.

AUI
Attachment Unit Interface. The interface between the coupler and transceiver.

Channel
Physical output of the Backplane.

Client
Equipment that initiates the communication.

Coupler
Communication interface.

CSMA/CD
Carrier Sense Multiple Access with Collision Detect. Multiple access with carrier and collision detection.

DATABASE
Database of 32 kbits and 32 kwords

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EPROM

Electronic component installed on the Backplane, on which non-erasable programs are stored.

ICMP
Internet Control Message Protocol. Control message protocol using IP.

IP
Internet Protocol. Inter-network protocol.

Item
Defines a unit article or element.

ITP
Industrial Twisted Pair. Double industrial twisted pair, developed by Siemens.

LLC
Logical Link Control.

OSI

Open Systems Interconnection.

TCP Port
Number associated with the IP address. It enables TCP to identify the application using the connection.

RFC
Request For Comment. Set of specifications describing all TCP/IP protocols.

TCP
Transmission Control Protocol.

UDP
User Datagram Protocol. Simple packet transmission protocol, with no connection or retry.

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